

torch.logsumexp#

<https://pytorch.org/docs/stable/generated/torch.logsumexp.html>

Description:

Returns the log of summed exponentials of each row of the input tensor in the given dimension `dim`. The computation is numerically stabilized. For summation index `jjj` given by `dim` and other indices `iii`, the result is $\log(\sum_j \exp(x_{iii, jjj}))$. If `keepdim` is `True`, the output tensor is of the same size as input except in the dimension(s) `dim` where it is of size 1. Otherwise, `dim` is squeezed (see `torch.squeeze()`), resulting in the output tensor having 1 (or `len(dim)`) fewer dimension(s). `input` (Tensor) – the input tensor. `dim` (int or tuple of ints) – the dimension or dimensions to reduce.

Function Signature

```
torch.logsumexp(input, dim, keepdim=False, *, out=None)#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.randn(3, 3)
>>> torch.logsumexp(a, 1)
tensor([1.4907, 1.0593, 1.5696])
>>> torch.dist(torch.logsumexp(a, 1),
torch.log(torch.sum(torch.exp(a), 1)))
tensor(1.6859e-07)
```

Detailed Documentation

Documentation

Returns the log of summed exponentials of each row of the input tensor in the given dimension `dim`. The computation is numerically stabilized.

For summation index `jjj` given by `dim` and other indices `iii`, the result is

If `keepdim` is `True`, the output tensor is of the same size as input except in the dimension(s) `dim` where it is of size 1. Otherwise, `dim` is squeezed (see `torch.squeeze()`), resulting in the output tensor having 1 (or `len(dim)`) fewer dimension(s).

`input` (Tensor) – the input tensor.

`dim` (int or tuple of ints) – the dimension or dimensions to reduce.

`keepdim` (bool, optional) – whether the output tensor has `dim` retained or not. Default: `False`.

`out` (Tensor, optional) – the output tensor.